Fire Extinguisher

Category: **Safety & PPE** Type: **Full SOP** Priority: **High** 60 labs

Covers these lab items: Fire extinguisher; Fire Extinguisher (ABC Type); Fire Extinguisher CO2; Fire extinguishers; Fire extinguishers and warning signs; Fire extinguishers, emergency exit maps, and signage; Fire extinguishers, first aid boxes, and workplace emergency signage; Fire Safety Equipment; Workshop Safety: Fire e

What it is used for

A fire extinguisher helps control small, early-stage fires when it is safe to act. Students and teachers must identify the correct extinguisher type and know when to evacuate instead.

Main parts

- Cylinder containing the fire-extinguishing agent under pressure.
- Safety pin prevents accidental discharge.
- Tamper seal indicates whether the extinguisher may have been used.
- Handle and operating lever release the extinguishing agent.
- Hose or nozzle directs the agent toward the fire.
- Pressure gauge indicates operating pressure on applicable models.
- Instruction label identifies fire classes and operating directions.

Before you start

1. Identify the nearest fire extinguisher and emergency exit before starting practical work.
2. Check the instruction label and confirm the extinguisher suits the fire type.
3. Inspect the cylinder, hose, nozzle, pin, seal, and body for damage.
4. Check the pressure gauge is in the operable zone, if fitted.
5. Ensure the access route is clear and the extinguisher is securely mounted.
6. Raise the alarm and call emergency services when a fire is discovered.

How to use

1. Raise the alarm immediately and inform the teacher or responsible staff member.
2. Assess the fire from a safe distance and identify the burning material.
3. Use an extinguisher only for a small, contained fire with a clear escape route.
4. Select the correct extinguisher; never use water on electrical or flammable-liquid fires.
5. Remove the safety pin while keeping the nozzle pointed away from people.
6. Stand at a safe distance and aim the nozzle at the fire's base.
7. Squeeze the operating lever to discharge the extinguishing agent.
8. Sweep the nozzle across the base of the fire until it is controlled.
9. Back away immediately if flames spread, smoke increases, or the extinguisher becomes ineffective.
10. Evacuate through the safe exit and do not re-enter until authorities permit.

⚠ Safety

- Never fight a large, spreading, or smoke-filled fire; evacuate immediately.
- Keep your back toward a clear escape route while using an extinguisher.
- Never touch electrical equipment or approach live electrical hazards during a fire.
- Use CO2 extinguishers carefully in confined spaces because they can reduce breathable oxygen.
- Never direct the discharge toward a person's face or bare skin.
- Only trained and authorised people should operate extinguishers during practical demonstrations.

Care & storage

- Keep extinguishers visible, accessible, securely mounted, and free from obstruction.
- Arrange scheduled inspection and servicing by authorised personnel according to the label and applicable requirements.
- Report damaged seals, low pressure, corrosion, leakage, or missing safety pins immediately.
- After any discharge, arrange prompt professional inspection, recharging, or replacement before reuse.

Common mistakes to avoid

- Choosing an extinguisher without checking the fire type and instruction label.
- Aiming at the flames instead of the base of the fire.
- Attempting to fight a fire after the escape route becomes unsafe.
- Returning to a fire area before authorised personnel declare it safe.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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